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*EE/CprE/SE 491 WEEKLY REPORT 3*

*9/30/26 – 10/6/26*

*Group number: sdmay27-18*

*Project title: Eddy Current (EC) Probes for Conductivity Measurements*

*Client &/Advisor:*

*Advisor – Jiming Song*

*Client – Dr. Mingyang Lu, CNDE and ISU-ECPE*

*Team Members/Role:*

*Connor Crimmins - Client and advisor communication.*

*Jonathan Gatewood - Software and data-processing coordination.*

*Eric Williams - Circuit and measurement electronics coordination.*

*Truong Luu - Team organization, scheduling, and task tracking.*

*Sebastian De La Torr- Proband sensing hardware coordination.*

*Benjamin Ponce-Martinez - Testing, calibration, and documentation coordination.*

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For this week the group put more time into our research of the project as well as communicated with Professor Song. We began our week of research by talking to Professor Song about Eddy Current and what the professor is looking for in this project. During the meeting we found out that eddy current is the current produced by a magnetic field that comes close in contact to a conductive material. The scope of this project is to design a probe that when it comes in contact with a metal it can read its conductivity based from the Eddy Current reading. After meeting with Professor Song we shared some of our thoughts with him for the project and left with intel that we should look into the NDE website and learn more before we go into building a probe. For this project the group split up the work into, learning about NDE, VNA searching, and designing of the project. The objective for the week was to learn more about NDE to understand the field more and then begin researching components. This week we completed tasks of looking into VNA, Coils, NDE, and minor designing of the project. The only changes that were made to the project is we cannot confirm now if we can use the AD5933 chip and should use other chips.

- Eric Williams: Eric looked into the possibility of using the AD5933 chip but found that it does not fit the scope of the project. He found that a more effective way to derive and calculate important values would be on a coding platform in the laptop. This way a GUI could be built where commands and important data are shown. Some possible platforms he looked into are Arduino IDE, VS Code, and Python.
- Connor Crimmins: Connor read more into NDE to understand more about eddy current. He found that the website includes projects similar to this that use circuitry to measure. He thinks this can be useful to design our project. He also made a design for the project that might be good for the remainder of the scope of our work. It is a handheld device that we can use to measure the conductivity. It uses a PCB that will have a memory chip and a measurement chip and has a coil for the probe. There will be a LCD screen that with the help of coding it will show our readings for conductivity. This is all thought for the future of this project.
- Jonathan Gatewood: Jonathan developed the calculation and modeling framework to support VNA measurements and conductivity calibration. This included skin depth, complex impedance, frequency response, and the expected change in the probe signal when the coil is placed near conductive material. He provided equations and example calculations to later be implemented in the data-processing software.
- Truong Luu: Last week, he looked into some factors that might affect the probe response apart from conductivity. He figured out that these factors are frequency, permeability, and distance between the coil and the metal.
- Sebastian De La Torre: After talking to the professor, Sebastian has put aside any sort of testing and conducted more research into probe and sensor fronted configurations.
- Benjamin Ponce-Martinez: Benjamin did some research on the materials that could be used to build the probe and looked into which metals could be used to test the probe.

| <b><u>NAME</u></b>      | <b><u>Individual Contributions</u></b><br><i>(Quick list of contributions. This should be short.)</i>   | <b><u>Hours this week</u></b> | <b><u>HOURS cumulative</u></b> |
|-------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------|
| Connor Crimmins         | Looked into NDE website to learn more about the theory and designed a baseline product for the project. | 6                             | 14                             |
| Jonathan Gatewood       | VNA calculation & modeling Framework                                                                    | 7                             | 15                             |
| Eric Williams           | Looked into AD5933 chip use in project. Researched possible coding platforms that could be used.        | 6                             | 14                             |
| Truong Luu              | Looked into the factor affecting the probe response.                                                    | 6                             | 14                             |
| Sebastian De La Torre   | Researched probe configurations and sensor requirements                                                 | 6                             | 14                             |
| Benjamin Ponce-Martinez | Researched metals that could be used to test the probe                                                  | 6.5                           | 15                             |

- Jonathan Gatewood: While we wait on materials and the design of things, Jonathan will move from the calculation framework into helping Eric look for coding platforms and research software techniques involved with the project with the limited software knowledge we have.
- Eric Williams: Eric will continue looking at possible coding platforms that can be used to develop a program for the calculations and display of data. The calculations for each parameter derived by Jonathan will be used in the code.
- Connor Crimmins: My goals are to learn more about the theory of this project as well as help with searching for good probe material and VNA chip.
- Sebastian De La Torre: Sebastian will continue to research more into probe configurations and sensor requirements and will be communicating with the advisor for feedback.
- Benjamin Ponce-Martinez: Benjamin will continue to research more materials that could be used to build the probe.
- Truong Luu: From looking into the factors that might affect the probe response, he will try to figure out what the right frequency and what the ideal distance between the coil and the metal need to be in order to receive the good response.

During this week, we had our first advisory meeting with both of our advisors on Wednesday. In this meeting, we were given a small lecture on the various probe types used in VNA calculations and a website where we could get valuable information on the physics, techniques, and engineering knowledge needed to do nondestructive evaluation efficiently. We asked several questions on the scope of the project and were able to better define our goal at the end.